



Introduction

Wearable technology has traditionally prioritized data display and numerical quantification, using the body’s physiological data to display and communicate internal information rather than personal self expression, while fashion couture has relatively stayed separate from technology. This project explores biomimetic kinetic fashion, merging technology, biology, and design into a single expressive system.

Mechanized 3D printed flowers open and close at a speed that is based on the wearer's heart rate. By translating physiological data into visible and blooming expression, the garment creates a beautiful and personal display of internal emotional and physical states.

Purpose

The purpose of this research is to design and create a fully wearable dress that displays the wearer’s inner physiological state by visually actuating a gentle, mechanical motion.

Limitations

- Lack of familiarity with CAD softwares
- MAX30102 sensor accuracy
- Single-User testing
- Mechanism durability
- Availability of 3D printing stations

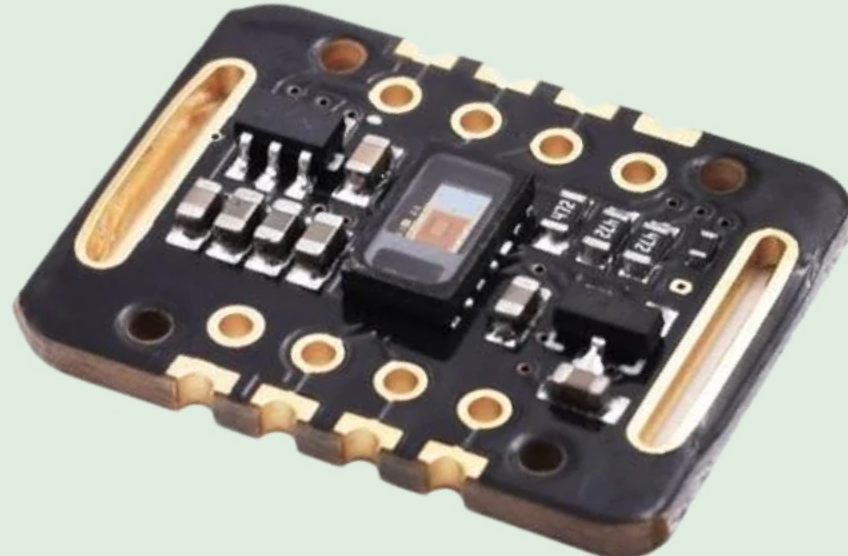


Methods

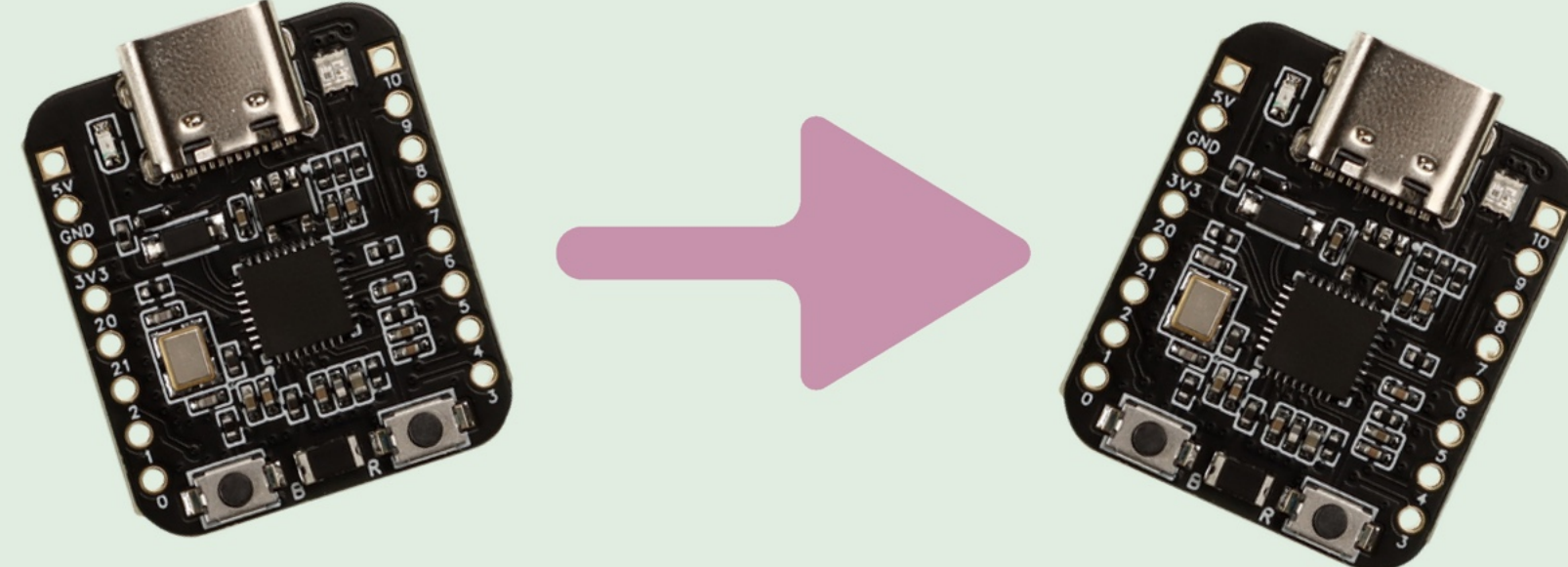
Mechanical System:
A servo actuates 3D printed petals mimicking flower blooms.



Sensing System:
MAX30102 measuring beats per minute in real-time, taking a scaled average to determine the rate of opening/closing.



Control System:
One sender ESP-32 takes the data from the MAX30102, and transmits it to another ESP-32 via ESPNOW, which processes heart rate data and mapping it to motor control signals.



Results

HR Avg: 107 BPM	Bloom Speed: 1.63 deg/frame	Servo: 10°	[CLOSING]
HR Avg: 108 BPM	Bloom Speed: 1.66 deg/frame	Servo: 49°	[OPENING]
HR Avg: 107 BPM	Bloom Speed: 1.63 deg/frame	Servo: 111°	[OPENING]
HR Avg: 106 BPM	Bloom Speed: 1.60 deg/frame	Servo: 75°	[CLOSING]
HR Avg: 105 BPM	Bloom Speed: 1.57 deg/frame	Servo: 17°	[CLOSING]
HR Avg: 103 BPM	Bloom Speed: 1.51 deg/frame	Servo: 42°	[OPENING]
HR Avg: 102 BPM	Bloom Speed: 1.49 deg/frame	Servo: 93°	[OPENING]
HR Avg: 103 BPM	Bloom Speed: 1.51 deg/frame	Servo: 94°	[CLOSING]
HR Avg: 106 BPM	Bloom Speed: 1.60 deg/frame	Servo: 39°	[CLOSING]
HR Avg: 107 BPM	Bloom Speed: 1.63 deg/frame	Servo: 16°	[OPENING]
HR Avg: 107 BPM	Bloom Speed: 1.63 deg/frame	Servo: 69°	[OPENING]
HR Avg: 108 BPM	Bloom Speed: 1.66 deg/frame	Servo: 113°	[CLOSING]
HR Avg: 109 BPM	Bloom Speed: 1.68 deg/frame	Servo: 52°	[CLOSING]
HR Avg: 110 BPM	Bloom Speed: 1.71 deg/frame	Servo: 10°	[OPENING]
HR Avg: 106 BPM	Bloom Speed: 1.60 deg/frame	Servo: 66°	[OPENING]
HR Avg: 103 BPM	Bloom Speed: 1.51 deg/frame	Servo: 116°	[CLOSING]
HR Avg: 100 BPM	Bloom Speed: 1.43 deg/frame	Servo: 61°	[CLOSING]
HR Avg: 96 BPM	Bloom Speed: 1.32 deg/frame	Servo: 11°	[CLOSING]
HR Avg: 96 BPM	Bloom Speed: 1.32 deg/frame	Servo: 34°	[OPENING]
HR Avg: 96 BPM	Bloom Speed: 1.32 deg/frame	Servo: 79°	[OPENING]
HR Avg: 101 BPM	Bloom Speed: 1.46 deg/frame	Servo: 111°	[CLOSING]
HR Avg: 102 BPM	Bloom Speed: 1.49 deg/frame	Servo: 49°	[CLOSING]
HR Avg: 106 BPM	Bloom Speed: 1.60 deg/frame	Servo: 0°	[CLOSING]
HR Avg: 108 BPM	Bloom Speed: 1.66 deg/frame	Servo: 65°	[OPENING]
HR Avg: 108 BPM	Bloom Speed: 1.66 deg/frame	Servo: 116°	[CLOSING]
HR Avg: 111 BPM	Bloom Speed: 1.74 deg/frame	Servo: 57°	[CLOSING]
HR Avg: 112 BPM	Bloom Speed: 1.77 deg/frame	Servo: 3°	[OPENING]
HR Avg: 111 BPM	Bloom Speed: 1.74 deg/frame	Servo: 67°	[OPENING]
HR Avg: 115 BPM	Bloom Speed: 1.85 deg/frame	Servo: 105°	[CLOSING]
HR Avg: 117 BPM	Bloom Speed: 1.91 deg/frame	Servo: 36°	[CLOSING]
HR Avg: 121 BPM	Bloom Speed: 2.02 deg/frame	Servo: 40°	[OPENING]
HR Avg: 112 BPM	Bloom Speed: 1.77 deg/frame	Servo: 114°	[OPENING]

Figure 1: Serial Output during Heart Rate Testing
Image created by researcher using Arduino, 2026.

Heart rate mapping: Resting (60-80 BPM) = slow blooms, Elevated (100-120 BPM) = rapid movements

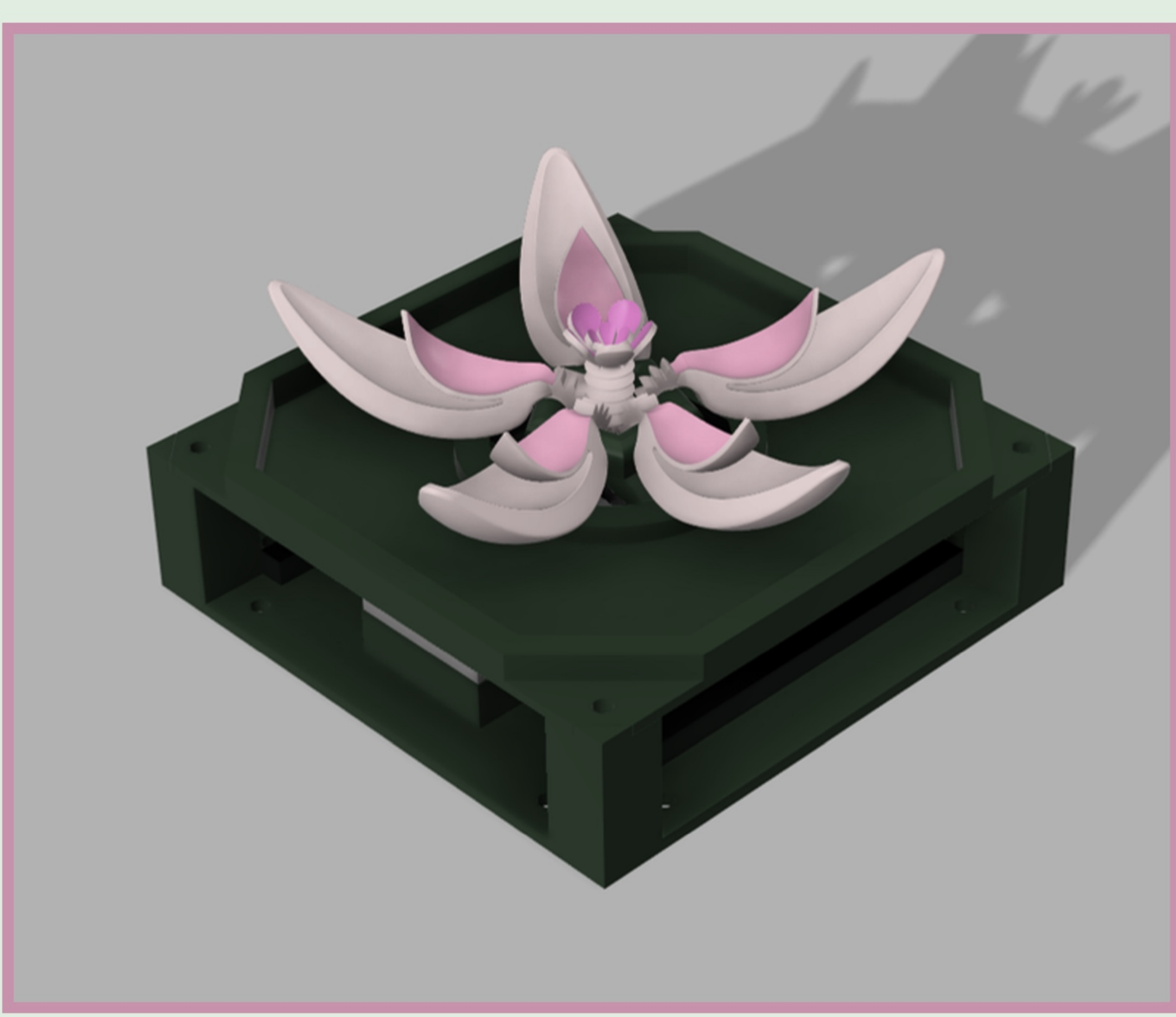


Figure 2: Render of Flower Mechanism
Image created by researcher using Fusion, 2026.

Mechanical Actuator: A rack and pinion system with a custom servo horn and geared stem move the petals.



Conclusion

This prototype demonstrates that physiological data can be effectively translated into kinetic display using biomimetic design. The flower mechanism successfully completed 203 cycles before being limited by servo overload, proving the sensor-to-actuator concept works.

However, limitations in time, printer availability, and software familiarity impacted the project timeline. The bodice structure has been digitally modeled but fabrication remains incomplete. Despite this, the core engineering principles and heart rate integration have been validated. Future work will focus on completing the bodice construction and refining integration.

Future Research

- Power optimization:** Explore battery options for running without a computer for a wearable amount of time.
- Design refinement:** Develop and engineer a wearable bodice primarily for comfort and aesthetics.

References

Blender (n.d.). Blender Logo [Image]. *Blender*: <https://www.blender.org/about/logos/>
Wikipedia (n.d.). Fusion360 Logo [Image]. *Wikipedia*: https://en.wikipedia.org/wiki/File:Fusion360_Logosvg
Voltaat (n.d.). Product Image of MAX30102 [Image]. *Voltaat*: <https://www.voltaat.com/products/pulse-oximeter-and-heart-rate-sensor-max30102>
Elecrow (n.d.). Product Image of ESP32-C3 [Image]. *Elecrow*: <https://www.elecrow.com/esp32-c3-mini-development-board-esp32-development-board-with-wi-fi-and-bluetooth.html>
Union Studio (n.d.). Pink Floral Watercolor Illustration [Graphic]. In *Canva Pro*: www.canva.com
Ernes, S. (n.d.). Watercolor Illustration of Pink and Yellow Flowers [Graphic]. In *Canva Pro*: www.canva.com
Presentation designed using Canva